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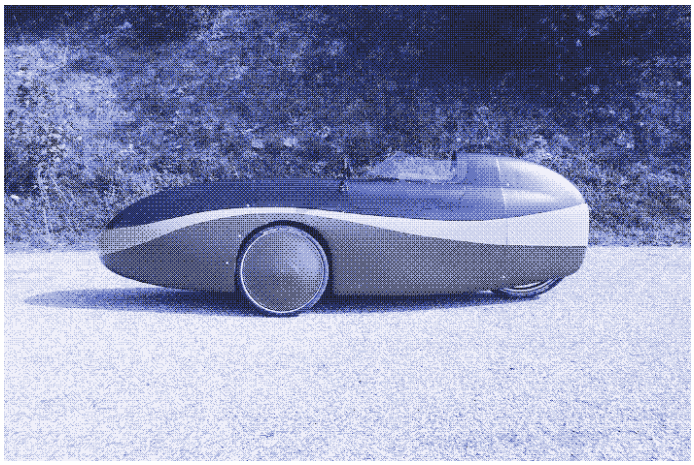
Electric Velomobiles: as Fast and Comfortable as Automobiles, but 80 times more Efficient

About a quarter of the existent wind turbines would suffice to power as many electric velomobiles as there are people.

October 24, 2012 Written by Kris De Decker Translations fr nl es pl



An electric velomobile. All pictures: Fietser.be (https://sites.google.com/site/mobilitylabbe/Home)

Both the velomobile and the electric bicycle increase the limited range of the cyclist – the former optimises aerodynamics and ergonomics, while the latter assists muscle power with an electric motor fuelled by a battery.

The electric velomobile combines both approaches, and so maximises the range of the cyclist – so much so that it is able to replace most, if not all, automobile trips.

While electric velomobiles have a speed and range that is comparable to that of electric cars, they are up to 80 times more efficient. About a quarter of the existent wind turbines would suffice to power as many electric velomobiles as there are people.

Few people find the bicycle useful for distances longer than 5 km (3 miles). In the USA, for instance, 85 % of bicycle trips involve a trip of less than 5 km. Even in the Netherlands, the most bicycle-friendly country in the western world, 77 % of bike trips are less than 5 km. Only 1 % of Dutch bicycle trips are more than 15 km (9 miles). In contrast, the average car trip amounts to 15.5 km in the USA and 16.5 km in the Netherlands, with the average trip to work being 19.5 km in the USA and 22 km in the Netherlands. (Sources: 1 (http://blog.bikeleague.org/blog/2010/01/national-household-travel-survey-short-trips-analysis/) , 2 (http://www.cbs.nl/nl-NL/menu/themas/verkeer-vervoer/publicaties/artikelen/archief/2008/2008-2539-wm.htm) , 3 (http://nhts.ornl.gov/2009/pub/stt.pdf) , 4 (http://nhts.ornl.gov/2009/pub/stt.pdf) , 5 (http://www.cvs-congres.nl/cvspdfdocs/cvs10_027.pdf) .)

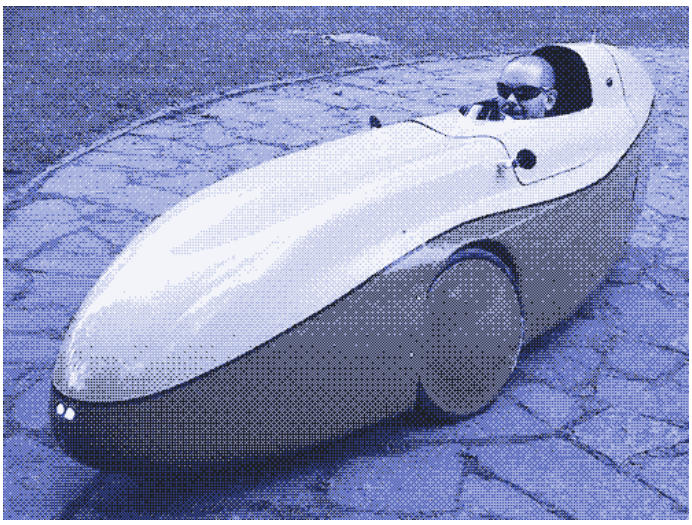
It's clear that the bicycle is not a viable alternative to the car. Depending on his or her fitness, a cyclist reaches cruising speeds of 10 to 25 km/h, which means that the average trip to work would take at least two to four hours, there and back. A strong headwind would make it even longer, and when the cyclist is in a hurry or has to climb hills, he or she would arrive all sweaty. When it rains, the cyclist arrives soaking wet, and when it's cold hands and feet would freeze. Longer trips on a bicycle also affect the body: wrists, back, shoulders and crotch all suffer, especially when you choose a faster bike.

An electrically-assisted bicycle solves some of these problems, but not all. The electric motor can be used to reach a destination faster, or with less effort, but the cyclist remains unprotected from the weather. Longer trips would still cause discomfort. Moreover, the range of most electric bicycles (about 25 km or 15.5 miles) is just large enough for the average one-way trip to work, which means that it will not suffice for all commutes.

The Advantages of an Electric assist Velomobile

The velomobile—a recumbent tricycle with aerodynamic bodywork (https://solar.lowtechmagazine.com/2010/09/the-velomobile-high-tech-bike-or-low-tech-car/) – offers a more interesting alternative to the bicycle for longer trips. The bodywork protects the driver (and luggage) from the weather, while the comfortable recumbent seat eases the strain on the body, making it possible to take longer trips without discomfort.

Furthermore, a velomobile (even without electric assistance) is much faster than an electric bicycle.



At speeds below 10 km/h (6 mph), rolling resistance is the biggest challenge for a cyclist. Air resistance becomes increasingly influential at higher speeds, and becomes the dominant force at speeds above 25 km/h (15.5 mph). This is because rolling resistance increases in proportion to speed, while air drag increases with the *square* of speed. Because a velomobilist has much better aerodynamics than a bicyclist—the drag coefficient of a velomobilist is up to 30 times lower—he or she can attain higher speeds with the same effort.

If rigged with an electric auxiliary motor, the weak points of the velomobile—its slower acceleration and climbing speed—are eliminated

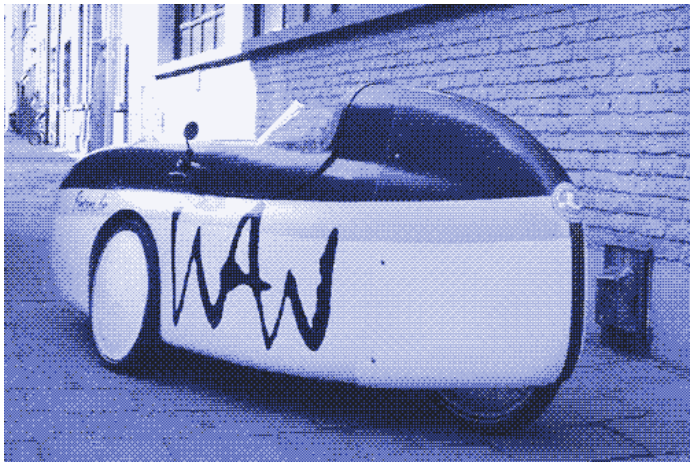
On the downside, a velomobile is heavier than a bicycle, which means that it takes more effort to accelerate and to climb hills. Acceleration is inversely proportional to the mass of a vehicle, so a velomobile uses roughly twice as much energy during acceleration than a bicycle, depending on the weight of the driver and vehicle.

If rigged with an electric auxiliary motor, the weak points of the velomobile—its slower acceleration and climbing speed—are eliminated. At the same time, a motor accentuates its advantages by further improving on the range of a cyclist. Last but not least, a battery will give a much better range in the velomobile, due to its better aerodynamics.

Test Driving a Ferrari

In August, I test drove an electrical velomobile—the eWAW (https://sites.google.com/site/mobilitylabbe/blog/e-wawabio-electrichybrid), a vehicle that is sold by Fietser.be (https://sites.google.com/site/mobilitylabbe/Home)—in and around Ghent, Belgium. Brecht Vandeputte, the driving force behind the Belgian manufacturer, accompanied me in an unassisted WAW during a one and a half hour trip through the city and along the tow path of the river Schelde.

The WAW velomobile (https://sites.google.com/site/mobilitylabbe/Home/waw/what-is-the-WAW/berichtzondertitel-1) (without electrical assistance) was originally developed for winning human-powered vehicle races. It was adapted for daily use with the addition of, among other things, a leakproof rear tyre, open wheel arches (which make the vehicle more agile), an adjustable seat, and a more durable body—which consists of a carbon roll bar and safety cage surrounded by aramid crumple zones. The WAW is known worldwide, at least among velomobilists, as one of the fastest velomobiles available on the market—some call it the *Ferrari* of the velomobiles.



The WAW stands out because of its weight (it is 28 kg, as opposed to 34 kg, the weight of the most popular velomobiles, the Dutch Quest (http://en.velomobiel.nl/quest/) and Alleweder (http://alleweder.jp-web.de/index.php?page=introduction&lang=en)) and its low centre of gravity (it has a ground clearance of only 9 cm and a height of just 90 cm). Along with a wide wheelbase, a hard suspension, and precise steering (it uses two gear sticks instead of one), this results in high speeds and excellent handling, even on sharp corners. Of course, the WAW also has the drawbacks you can expect from a real sports car, like the very basic interior finish and the fact that the vehicle rattles like a box of rocks when you ride it over a cobblestone road. If road conditions are bad, other

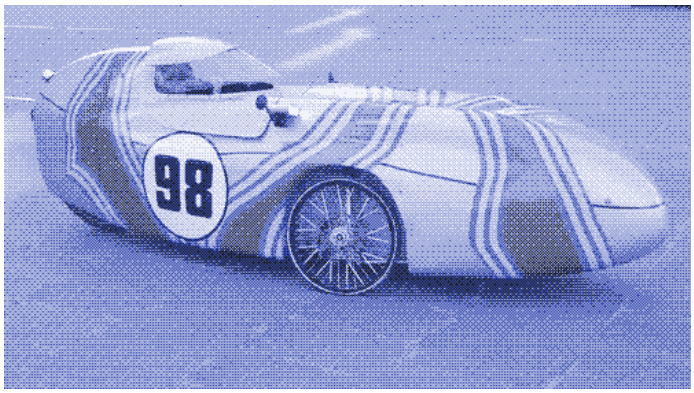
velomobiles with more comfortable suspension will be a better choice.

> With 250 watts of power, the electric motor of the eWAW gives a person with an average fitness level the power output of an athlete

The eWAW that I drove has everything that the WAW has, plus an electric motor of 250 watts and a surprisingly small battery of 288 Wh, which takes you 60 to 130 km further (37 to 81 miles). The battery and the motor add only 5 kg, bringing the total weight of the vehicle to 33 kg. This is comparable to the weight of other velomobiles without electric assistance. Hence, this pedal powered Ferrari is more than 10 kg lighter than other velomobiles, with a 250 watt electric assistance, such as the hybrid Alleweder and the e-Sunrider, which weigh 45 kg.

Cycling at 50 km/h

So how fast is the WAW, and how much faster is the eWAW? First of all, the eWAW is a hybrid vehicle, but the biomass powered motor, also known as the driver, is not included in the package. Because the driver always provides the main part of the total power output, the speed of the vehicle will depend on the power that he or she can deliver. There is no better illustration of this than my test drive. Over a period of about an hour and a half, Brecht and I managed to reach an average speed of 40 km/h (25 mph)—I was in the eWAW and had the regular assistance of the electric motor, Brecht was in a WAW *without* pedal assistance.



Cycling literature makes a distinction between three types of cyclists: people with an average fitness level, people with a good fitness level, and top athletes. Riders with an average fitness can maintain a power output of 100 to 150 watts over a period of one hour. Riding a WAW, this translates to speeds of 35 to 40 km/h in ideal conditions—an unobstructed racetrack, and a completely closed vehicle. Drivers with a good fitness level can deliver 200 watts of power over a period of one hour, which translates to speeds of 45 to 50 km/h under the same circumstances.

With 250 watts of power, the electric motor of the eWAW gives a person with an average fitness level (like me) the power output of an athlete (100 + 250 watts = 350 watts).

Maximizing Range and Efficiency

I am a speed freak, so when I found myself on a nice, open stretch of road, the first thing I did was start the motor at full throttle and pedal like a madman at the same time. If I could have more than 350 watts at my disposal, I calculated, I must be able to reach speeds of at least 70 or 80 km/h (40 to 50 mph). However, my attempt to go any faster than 50 km/h (30 mph) left me frustrated—the vehicle lacks the high gears needed for those speeds.

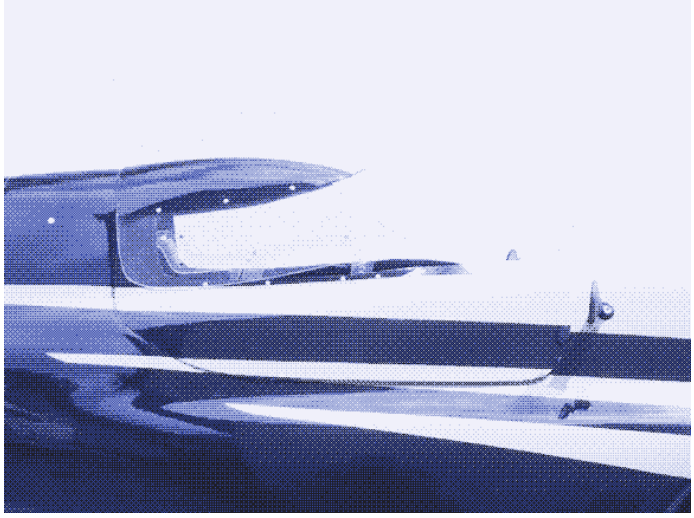
Why? Because the eWAW is designed for maximum efficiency. The electric motor is intended to be used for acceleration only (and for climbing hills). Once the velomobilist reaches a cruising speed of about 40 to 50 km/h, he or she switches to pedalling alone.

> The engineer's choice to assist the driver only during acceleration is smart; it increases the range of both the cyclist and the battery spectacularly

The eWAW does not increase the cruising speed or top speed of the unassisted WAW, although it does increase the average speed because it speeds acceleration. This is a different approach from the electric bicycle, where pedal assistance is continuous at normal cruising speeds. With regards to efficiency, the concept behind the eWAW makes much sense.

A bicyclist needs less energy to accelerate than a velomobilist does (because of the bike's lighter weight) but more energy to keep up speed (because of its weak aerodynamics). In contrast, a velomobilist needs more energy to accelerate than a bicyclist does (because of the

vehicle's heavier weight) but less energy to keep up speed (because of its excellent aerodynamics).



Because it takes more energy to accelerate in an eWAW than to drive it at a constant speed, the engineer's choice to assist the driver only during acceleration is smart; it increases the range of both the cyclist and the battery. The electric motor supports the driver during peak efforts, so that his or her endurance will increase spectacularly. (Peak efforts have a detrimental effect on endurance, while pedalling at a steady pace can be done for hours.) Meanwhile, the driver offers the same service to the battery. Because the electric motor is shut off at cruising speed, the battery range increases considerably.

This said, the driver of the eWAW can choose to use the motor at cruising speed, because it can be operated at his or her will by means of a throttle. This is how I drove the vehicle. As a consequence, the battery lasted 'only' 60 km (37 miles), but at least I could keep up with Brecht.

80 times More Efficient than Electric Cars

Mounting an electric engine in a velomobile is controversial among velomobilists, just as an electric bicycle is skewed by many biking aficionados. However, when we compare the eWAW with the electric car, still viewed by many as the future of sustainable transportation, it's a clear winner. In fact, the electric velomobile is everything what the electric car wants to be, but isn't (http://www.notechmagazine.com/2012/10/life-cycle-analysis-of-electric-cars.html) : a sustainable alternative to the automobile with combustion engine. It is nearly impossible to design a personal, motorised and practical vehicle that is more efficient than the eWAW.

> If all 300 million Americans replace their car with an electric velomobile, they need only 25 % of the electricity produced by existing American wind turbines

A simple calculation can illustrate this claim. Imagine that all 300 million Americans replace their car with an electric velomobile and all drive to work on the same day. To charge the 288 Wh battery of each of these 300 million eWAW's, we need 86,4 GWh of electricity. This is only 25 % of the electricity produced by existing American wind turbines (on average per day during the period July 2011 to June 2012, source (http://www.eia.gov/electricity/monthly/pdf/epm.pdf)). In other words, we could make a switch to private vehicles operating on 100 % renewable energy, using existent energy plants.

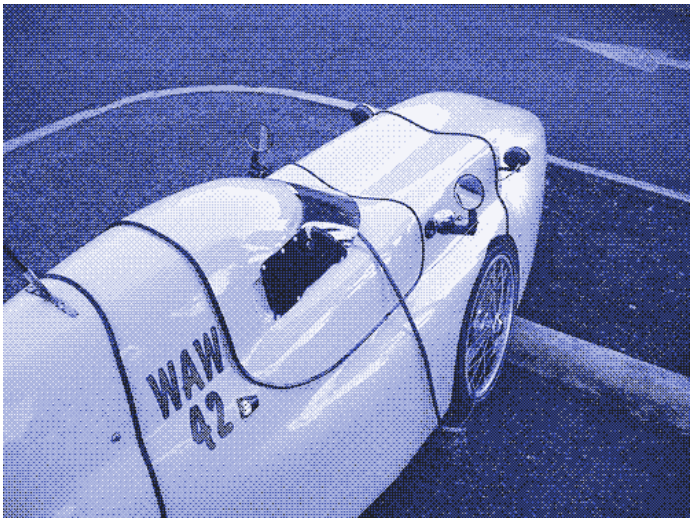


Photo credit: Bill Bates (http://www.flickr.com/wmbates/sets)

Now imagine that all 300 million Americans replaced their cars with an electric version like the Nissan Leaf, and all drive to work on the same day. To charge the 24 kW battery of each of those 300 million vehicles, we need 7,200 Gwh of electricity. This is *20 times more than what American wind turbines produce today*, and 80 times more than what electric velomobiles need. In short: scenario one is realistic, scenario two is not.

Even if we all started carpooling, and every electric automobile could carry five people, there remains a large gap in efficiency. Charging 60 million electric cars would still require 16.6 times more electricity than charging 300 million eWAW's. The electric velomobile also makes it fairly easy for a driver to charge his or her own vehicle. A solar panel of about 60 watts (with a surface area of less than

one square metre) produces enough energy to charge the battery, even on a dark winter day.

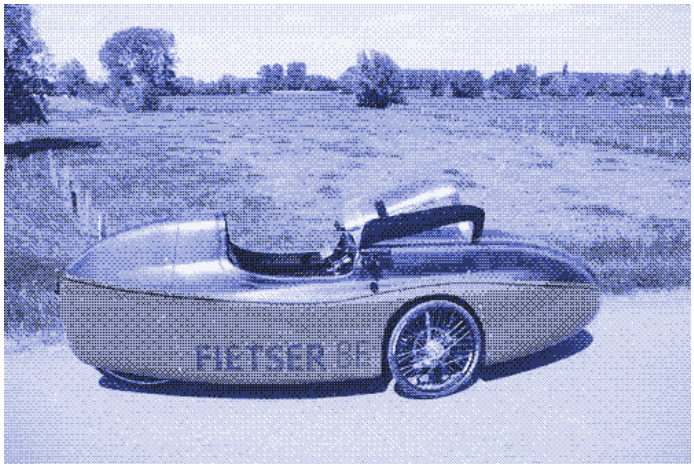
In Europe, it would take an even smaller share of the existent wind turbines to charge every European's eWAW. For the sake of thoroughness, it should be mentioned that the bio-motor also requires energy: the driver needs to eat, and this food needs to be produced. But since western people eat too much, and then drive their cars to the gym in order to lose excess fat, this factor can be safely ignored.

Range Anxiety

The large difference in energy efficiency between electric velomobiles and electric cars is remarkable, because both have a similar range. As mentioned, the eWAW takes you a distance of 60 to 130 km, depending on how intensively you use the motor. The Nissan Leaf takes you at best 160 km (https://solar.lowtechmagazine.com/2010/05/the-status-quo-of-electric-cars-better-batteries-same-range/) , when you drive slowly and steadily, and when you don't make use of the air-conditioning, heating or electronic gadgets on board.

> Adding only 6 kg of batteries increases the range of the electric velomobile to 450 km

A heating system is not required in a velomobile, not even in winter, because hands and feet are protected from the wind by the bodywork, and because the driver is active (body activity is the most important factor in maintaining thermal comfort (https://solar.lowtechmagazine.com/2011/02/insulation-first-the-body-then-the-home/)). The need for cooling in summer, on the other hand, will decrease the range—the driver will rely more on the electric motor in order to cool down.



Interestingly, it is easier to increase the range of the electric velomobile than of the electric car, if necessary. The eWAW can be equipped with one or two extra batteries, which increases the range up to 180 km (112 miles, with continuous assistance of the motor) or 450 km (280 miles, when the motor is only used to assist acceleration). Adding two batteries to the eWAW increases the weight of the vehicle by only 6 kg, and still leaves ample space for luggage. If we suppose that the rider weighs 70 kg, then adding two batteries increases the total weight of the eWAW from 103 to 109 kg—a weight gain of 6 %. If we apply the same trick to the Nissan Leaf (where three times as many batteries take the place of the rear seat and the trunk), total weight increases from 1,582 kg (the driver of 70 kg included) to 2,022 kg—a weight gain of 30 %.

Another way to increase the range of an electric vehicle is swapping batteries or fast-charging them. These options are available for both electric cars and velomobiles, but developing a charging infrastructure for electric cars is a daunting task, while doing so for electric velomobiles is easy. The battery of the eWAW not only needs 80 times less energy than the battery of a Nissan Leaf (which makes fast-charging (https://solar.lowtechmagazine.com/2009/03/who-killed-the-electric-grid-fast-charging-electric-cars/) a real option), it also weighs 73 times less (which makes swapping batteries a very low-tech operation). While we do have faster vehicles for long distances that are equally sustainable (like trains and trolleybusses (https://solar.lowtechmagazine.com/2009/07/get-wired-again-trolleybuses-and-trolleytrucks/) , the velomobile offers an alternative for those who prefer a personal means of transportation, or for those who prefer an active lifestyle.

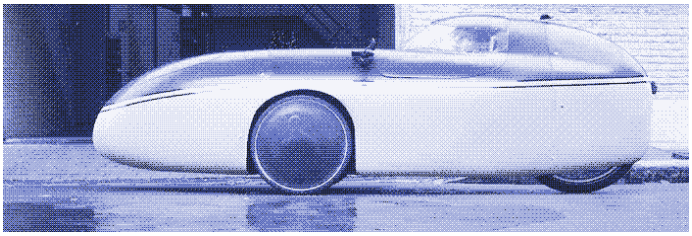
> The capacity of our roads would at least quadruple if we switched from cars to velomobiles

When the battery of an electric velomobile drains, the velomobilist can still pedal home—at speeds above those of a bicycle. The driver of the electric car can't do that, because his contraption is too heavy. One Nissan Leaf weighs as much as 46 eWAW's. Most of the energy used by an electric car (and by a car with combustion engine), is used to move the vehicle itself, not the driver—the Nissan Leaf is 21 times heavier than its driver. In the case of the eWAW, this relation is reversed: the driver weighs two to three times more than the vehicle.

Fast and Smooth Traffic

The eWAW makes cycling a fast and comfortable option for longer distances. At a cruising speed of 50 km/h (31 mph), the average commute in the USA (19.5 km or 12 miles) would take 23.4 minutes. This compares very favourably with the car, for which the average commute time is 22.8 minutes (source (http://nhts.ornl.gov/2009/pub/stt.pdf)). In the Netherlands, where road traffic is heavy, the electric velomobile is potentially faster than a car. The velomobile

could cover the average commute of 22 km (13.7 miles) in 26.4 minutes, while it takes 28 minutes by car (source (http://www.cvs-congres.nl/cvspdfdocs/cvs10_027.pdf)).



Of course, a cruising speed of 50 km/h does not mean that a velomobilist can reach an average speed of 50 km/h during the whole trip. If cars could maintain their maximum cruising speed during the commute, they would be much faster than velomobiles. In reality, however, they can't do that because of speed limits, traffic lights and traffic jams.

Velomobiles could suffer similar delays, but there is an important difference: because a velomobile occupies much less space than a car (one car needs as much space as four velomobiles), free-flowing traffic is a much more realistic option for velomobiles. The capacity of our roads would at least quadruple if we switched from cars to velomobiles. Furthermore, the cruising speed of a velomobile does not exceed most speed limits.

Pimp up your Velomobile

Over and above this, it is easy to equip a velomobile with a more powerful motor and higher gears, allowing for much higher cruising speeds. It would lose efficiency and range, but, since an eWAW is 80 times more efficient than an electric car, there is quite a bit of room for pimping up a velomobile. We'll discuss these possibilities, as well as the legal obstacles for electric velomobiles, in the second part of this article.

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